ProspectCV: LLM-based Advanced CV-JD Evaluation Platform

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2024 IEEE Students Conference on Engineering and Systems (SCES) | 979-8-3503-7471-1/24/$31.00 ©2024 IEEE | DOI: 10.1109/SCES61914.2024.10652548

***Abstract*— This paper presents a novel framework that utilizes Google's Gemini Pro Vision large language model (LLM) and natural language processing (NLP) techniques to analyze and compare resumes or CVs with job descriptions. The framework not only evaluates the compatibility between the resume and job description but also provides personalized feedback and suggestions for improvement. By identifying potential gaps in skills and qualifications, and recommending relevant resources, the framework assists both human resources professionals in streamlining candidate shortlisting and empowers candidates to enhance their CVs. Gemini processes a resume as a Base64-encoded string and cleans up the job description using NLP. It then generates a personalized report with insights, including a compatibility score, skills analysis, improvement suggestions, and resource recommendations. Initial results demonstrate the framework's effectiveness in accurately assessing candidate suitability and providing valuable feedback. Future work will focus on expanding the framework to analyze multiple resumes simultaneously and developing more sophisticated algorithms for enhanced accuracy and reliability. This framework has the potential to revolutionize the recruitment process by streamlining candidate selection, empowering candidates to improve their CVs, and ultimately leading to better candidate job matches and a more efficient and equitable hiring landscape.**

Keywords—large language models, Gemini pro, resume, job description, screening system, applicant tracking system.

I. Introduction

The recruitment process is a crucial phase for companies that seek to find the best candidates for a job opening. However, with the increasing number of job seekers, companies often receive a flood of resumes, making it difficult to quickly identify the most qualified candidates. To overcome this challenge, the use of automation has gained momentum, with artificial intelligence and machine learning algorithms coming to the forefront. These technologies have made it possible to extract relevant information from resumes with speed and accuracy, enabling recruiters to streamline the selection process. The process of automating resume screening involves analyzing resumes for specific sections, such as education, skills, and other important details. Once the relevant information is extracted, it can be used to match resumes with job descriptions based on set criteria. This not only saves recruiters' time but also enables them to focus on the most qualified candidates. With the rise of online resume platforms and e-recruitment, there is an overwhelming amount of data to sift through to find valuable insights and metadata. The sheer volume of resumes to screen can be daunting and time-consuming. However, implementing an automated framework is the key to optimizing this process. One of the best solutions for this purpose is to use Large Language Models (LLM), which is a highly reliable and increasingly popular approach.

This paper introduces an advanced framework that evaluates a user's CV or resume and compares it with a given job description. The framework uses Google's Gemini Pro Vision LLM to process the CV and job description and returns data in JSON format. The primary aim of this framework is not only to evaluate and match the user's CV with the job description but also to provide valuable feedback and suggestions on how to improve the CV. By analyzing the CV and comparing it with the job description, the framework identifies potential gaps and shortcomings in the user's CV, and provides recommendations on how to address them. Moreover, the framework goes beyond feedback and recommendations, as it also suggests resources that can help users enhance their skills and qualifications. By recommending relevant training courses or certifications, the framework can guide users towards building a stronger and more competitive CV. Hence, this framework bears potential for two-fold application; not only can it aid HR professionals in their task of candidate shortlisting, but it also offers a means for prospective candidates to evaluate their CV against a given job description.

II. Related Work

The job market is going through a major change with the emergence of Artificial Intelligence and Machine Learning. These technologies are becoming more prevalent in automating and enhancing different parts of the hiring process, especially in areas such as resume screening and job description matching. This survey delves into the current state of research in this domain, exploring the latest advancements, identifying research gaps, and discussing future directions. Traditionally, resume screening has been a manual and time-consuming process, often relying on keyword-based matching to identify potentially qualified candidates [1]. However, this approach can be inefficient and prone to human bias, potentially overlooking qualified candidates or perpetuating discriminatory practices. AI and ML offer a more sophisticated and data-driven approach to resume screening and job description matching. Several AI and ML techniques are employed for this purpose. One common approach utilizes TF-IDF and cosine similarity to measure the relevance of resumes to specific job descriptions [2]. TF-IDF, or Term Frequency-Inverse Document Frequency, is a statistical measure that reflects the importance of words in a document relative to a corpus. Cosine similarity then measures the similarity between the TF-IDF vectors of the resume and job description, providing a more nuanced assessment compared to simple keyword matching.

Beyond these techniques, machine learning models, such as Support Vector Machines and Random Forests, can be trained on labeled data to classify resumes as relevant or irrelevant for specific positions [3]. This supervised learning approach allows the models to learn from past data and make predictions on new resumes with improved accuracy. Deep learning models, such as Convolutional Neural Networks and Recurrent Neural Networks, offer even greater accuracy by automatically extracting features from resumes and job descriptions, eliminating the need for manual feature engineering [4]. Furthermore, large language models like BERT are demonstrating superior performance in understanding the complex semantic relationships between skills, experiences, and job requirements [5]. These models are trained on massive text datasets and can capture nuanced meanings and relationships that traditional methods may miss. This allows for a more comprehensive and accurate assessment of candidate suitability for specific roles.

The use of AI and ML in resume screening and job description matching offers several benefits. First, AI-powered systems can screen large volumes of resumes quickly and efficiently, significantly reducing the time and cost associated with manual screening [6]. This is particularly beneficial in high-volume hiring scenarios where manually reviewing each application is impractical. Second, AI models can analyze resumes and job descriptions with greater depth and accuracy compared to humans, leading to better candidate-job matching [7]. This can result in improved hiring outcomes and reduced employee turnover. Additionally, AI systems can be designed to mitigate human biases and promote diversity and inclusion in the hiring process [8]. By focusing on skills and qualifications rather than demographic factors, AI can help reduce unconscious bias and create a more equitable hiring process. Moreover, AI can analyze a candidate's skills, experience, and career aspirations to recommend relevant job openings, improving the candidate's experience and increasing the likelihood of finding a suitable job [9].

Despite the significant advantages, several research gaps and challenges remain. One key challenge is developing AI models that can accurately assess complex skills and experience requirements described in job descriptions [10]. Current systems often struggle with nuanced language and may overlook qualified candidates if their resumes do not precisely mirror the wording used in the job description. Additionally, most systems primarily focus on matching technical skills and experience, neglecting soft skills and personality traits that are crucial for job success. Integrating these factors into AI models remains a challenge. Another critical concern is data quality and bias. AI models are susceptible to biases present in the training data, which can lead to unfair and discriminatory outcomes. Ensuring data quality and mitigating bias is crucial for fair and effective recruitment. Furthermore, many AI models operate as "black boxes," making it difficult to understand the rationale behind their predictions. Explainable AI models are needed to ensure transparency and trust in the hiring process. Finally, the use of AI in recruitment raises ethical concerns regarding fairness, privacy, and potential discrimination. Further research is needed to develop ethical frameworks for AI-powered recruitment tools.

Research in AI and ML for resume screening and job description matching is rapidly evolving. Future directions include developing AI models that can understand complex skills and experience requirements, incorporating soft skills and personality traits into AI models, ensuring data quality and mitigating bias, developing explainable AI models for recruitment, and establishing ethical frameworks for AI-powered recruitment tools. In conclusion, AI and ML are transforming the way resumes are screened and matched with job descriptions. While significant progress has been made, research efforts must continue to address existing gaps and ensure the development of fair, transparent, and ethical AI-powered recruitment systems. By addressing these challenges and pursuing future research directions, AI and ML have the potential to revolutionize the recruitment process, leading to better candidate-job matches and a more efficient and equitable hiring landscape.

III. CONTRIBUTIONS

- The paper presents a novel framework that assesses a user's CV or resume and compares it with a given job description. The framework employs Google's Gemini Pro Vision LLM to analyze the CV and job description and returns data in JSON format.

- The framework has a dual application. Firstly, it can assist HR professionals in their candidate selection process, aiding them in shortlisting candidates. Secondly, it offers a means for candidates to assess their CV against a specific job description.

- The framework suggests resources that can assist users in improving their skills and qualifications. By recommending relevant training courses or certifications, the framework can guide users towards creating a more competitive CV.

IV. FLOWCHART

The flowcharts illustrated in Fig. 1 and Fig. 2 provide a comprehensive overview of the system. The process commences with the user uploading their resume and the job description. The system validates the authenticity of the files to ensure that it is working with the correct documents. Once validated, the resume is converted into an image format. This conversion is carried out to streamline processing for Gemini and enhance the understanding of the resume. The image is then encoded as a base64 string to facilitate its transfer as a JSON field. The base64 encoding preserves the details while enabling better transfer over the Gemini API. Meanwhile, the job description is prepared and formatted to optimize analysis.

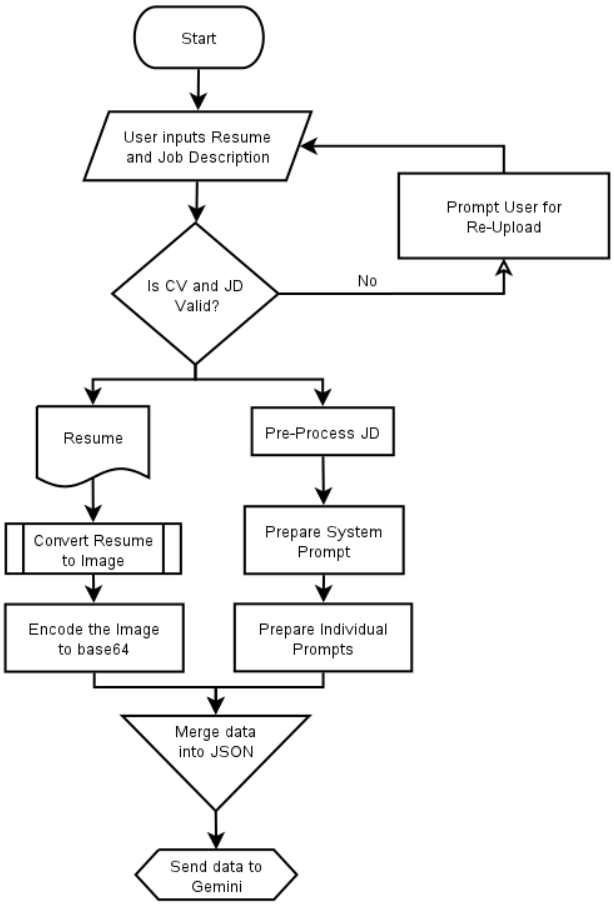


Fig. 1. Flow chart illustrating the process before sending data to Gemini Pro LLM.

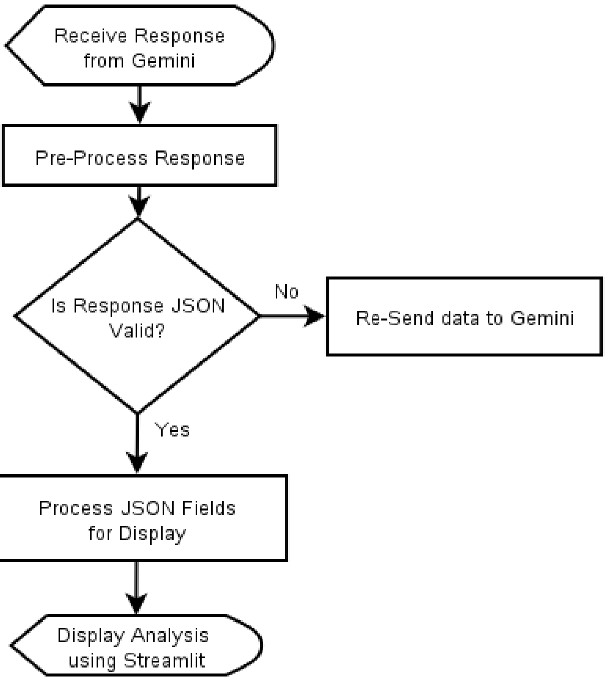


Fig. 2. Flow chart illustrating the process after sending data to Gemini Pro LLM.

Next, the data assembly stage is initiated, where the image-formatted resume and the processed job description are merged into a single JSON file. The use of JSON, being a lightweight and human-readable format, serves as a neat package for Gemini to comprehend the data it needs. A system prompt has been developed for this application alongside individual prompts for each function. The JSON file, together with the prompt module file, is then forwarded to Gemini, the star player of this project.

Gemini, with its analytical prowess, delves deep into both documents. It compares the skills and experiences highlighted in the resume with the requirements and desired qualifications outlined in the job description. Based on this comprehensive analysis, Gemini generates a personalized report packed with valuable insights. This report includes the candidate's basic information such as name, contact details, and years of experience. Additionally, it provides a thorough analysis and comparison of the user's skills and the skills required (missing skills) in the job description, a score of 0 to 100 based on the resume's compatibility with the job description, tips on tailoring the resume to match the job description better, highlighting relevant skills the candidate might have underplayed, or even suggesting areas in the resume that could benefit from strengthening. It also recommends resources like books and courses that the candidate can use to further improve their skills. These analyses are then displayed using the Streamlit library, which allows for an interactive user interface to view and better understand the generated analysis.

Leveraging Gemini's language processing capabilities assists HR professionals in their candidate selection process, aiding them in shortlisting candidates, and also empowers the candidates to refine their resume, showcase their qualifications more effectively, and ultimately increase their chances of landing that dream job.

V. Proposed Method

Import Libraries: Import the following libraries:

google.generativeai: The google.generativeai library in Python provides developers with access to advanced generative AI models developed by Google. The google.generativeai library is a convenient way for developers to harness the power of generative AI technology for a variety of applications.

pandas: The Pandas library is a versatile and powerful data analysis and manipulation tool for Python. Pandas encompasses a broad range of functionalities, for instance, it supports reading and writing data from a range of file formats such as CSV, Excel, JSON, and SQL databases.

tempfile: The Python tempfile library offers a simple way to create and handle temporary files and directories. By utilizing tempfile, developers can avoid the trouble of manually managing temporary files and guarantee that their programs leave no unnecessary files behind.

os: The os library in Python provides functions for interacting with the operating system. It offers a wide range of functions for interacting with the file system, managing processes, and accessing environment variables.

json: The json library in Python provides tools for working with JSON (JavaScript Object Notation) data. JSON is a lightweight, human-readable data format that is widely used for data exchange and storage. It provides a straightforward and efficient way to exchange data with other applications and services, making it a valuable tool for web development, data analysis, and other data-driven tasks.

streamlit: Streamlit is a powerful and user-friendly Python library for building interactive web applications. It simplifies the process of creating data-driven applications by providing a declarative and intuitive API. You can create complex and interactive applications with just a few lines of Python code. Streamlit handles the underlying web development complexities, allowing you to focus on your data and analysis.

Function to initialize Gemini Pro Vision: A function named get_gemini_response is initialized. It uses Google's generative AI model named "gemini-pro-vision" to generate content based on the provided inputs. Here's a breakdown of what it does:

Function Definition: The function get_gemini_response takes three arguments: input, pdf_content, and prompt. These represent different sources of information that can be used by the model to generate content.

Model Initialization: Inside the function, a Generative Model object named 'model' is created using genai (Generative Model constructor). The model name specified is "gemini-pro-vision". This model developed by Google is a pre-trained language model capable of generating text based on input data.

Input Preparation: A list comprehension is used to create a list called 'out'. This list will contain the non-null values from the input arguments (input, pdf_content[0], and prompt). This ensures that only valid inputs are passed to the model.

Content Generation: The model.generate_content() method is called with the out list as input. This method uses the generative model to process the provided inputs and generate new content. The result is stored in the response variable.

Resume pre-processing module: This module converts the resume from PDF to a list of images. Each image in the list represents a page from the resume. This conversion is carried out to streamline processing for Gemini and enhance the understanding of the resume. Once the image byte array is generated, it is further encoded into a Base64-encoded string, which facilitates its transfer as a JSON field. The use of base64 encoding ensures that the details of the original resume are preserved while enabling more efficient transfer over the Gemini API. Overall, this approach helps to streamline the processing of resumes and enhances the system's ability to extract relevant information from these documents.

Job Description pre-processing module: The pre-processing and cleansing of text data is an essential step in Natural Language Processing (NLP) to prepare raw text for analysis and modelling. This module pre-processes and cleans the Job Description text data. Various text pre-processing techniques have been used to enhance the quality and performance of the analysis. The text has been normalized to lowercase, and accents and special characters have been removed to ensure consistency in word representation. Tokenization of the text into individual words or meaningful units has been performed to facilitate further analysis at the word level. In addition, stop words have been removed, and punctuation has been eliminated to improve data quality. The handling of numbers and special characters has also been implemented. The elimination of noise and inconsistencies has resulted in improved accuracy and efficiency.

Prompt analysis module: We have implemented prompt engineering techniques to guide the model's output. A system prompt and individual prompts for each function are developed. The system prompt defines the model's role as a smart assistant for students and professionals seeking to enhance their resumes. It also specifies the desired style, tone, and limitations of the model, such as generating JSON content exclusively and avoiding content that is not inferred from the text. This system prompt remains effective for all content produced by the model, ensuring consistency throughout. Additionally, we have established a specific prompt for each individual function that provides relevant information, including profile information, resume analysis, improvements, and resources. This allows for customization and tailoring of content to specific resumes and job descriptions. By combining a persistent system prompt with tailored individual prompts, we have ensured that the generated content is not only consistent in style and tone but also relevant and effective for each resume and job description. This approach has helped generate high-quality content that is informative, engaging, and persuasive. As prompt engineering is an iterative process, we have continuously experimented with different prompts and evaluated the generated outputs to find the most effective approach for our use case.

Processing data with Gemini Pro LLM: The image-formatted resume and the processed job description are merged into a single JSON file. This is done to make the data easily comprehensible for Gemini, owing to the lightweight and human-readable nature of JSON. The JSON file, along with the prompt module file, is then passed on to Gemini Pro LLM. With its powerful analytical capabilities, Gemini meticulously scrutinizes both documents. It conducts a thorough comparison of the skills and experiences highlighted in the resume with the requirements and desired qualifications outlined in the job description. Based on this comprehensive analysis, Gemini generates a personalized report that is packed with valuable insights.

User Interface and output: The user interface of the application is developed using the streamlit library in python. Interactive widgets, such as sliders, buttons, and drop-down menus are used to make the application interactive. The analysis generated by Gemini Pro LLM is then presented using the Streamlit library. This enables the users to view and understand the generated analysis more effectively.

VI. Results And Observations

The initial step involves the user being prompted to upload both their resume and the job description. After which, the user is required to click on the submit and process button.

Gemini, with its strong analytical capabilities, performs a thorough examination of both documents. It conducts a detailed comparison of the skills and experiences highlighted in the resume with the requirements and desired qualifications outlined in the job description. Based on this comprehensive analysis, Gemini generates a personalized report that is packed with valuable insights. Upon successful processing, the resulting output is presented in four distinct tabs, as depicted in Fig. 3.

Profile Information | Resume Analysis | Improvements | Resources

Fig. 3. Result tabs.

Tab 1 Profile Information: The Profile Information Tab affords a comprehensive overview of the candidate's basic details, such as their name, contact information and experience. As seen in Fig. 4, the application displays a summary of the candidate's basic information, including their name, contact information, website, and location. Additionally, the Gemini Pro LLM infers and calculates certain aggregate functions, such as the candidate's experience in years, which may not be explicitly stated in the resume.

Profile Information:

Name: Robert Smith

Email: info@robertsmith.com

Phone: (123) 456 78 99

Website: www.robertsmith.com

Location: Alabama

Work Experience (In Years): 6

Fig. 4. Profile information.

Tab 2 Resume Analysis: The Resume Analysis Tab furnishes a detailed analysis and comparison of the candidate's skills and the skills demanded by the job description. It also provides a score, ranging from 0 to 100, based on the level of compatibility between the resume and the job description.

Case 1:

Candidate Resume: Android Developer.

Job Description Role: Android Application Developer.

As seen in Fig. 5, a compatibility score of 93% is shown between the uploaded resume and the job description. The score is high as both the resume and job description are for the role of an Android Developer; hence the match is strong. A detailed analysis and comparison of the candidate's skills and the skills demanded by the job description is also displayed as seen in Fig. 5.

Case 2:

Candidate Resume: Data Scientist.

Job Description Role: Android Application Developer.

As seen in Fig. 6, the uploaded resume has a compatibility score of only 30% with the job description. This low score is due to the fact that the resume and job description are for different roles and do not match. Additionally, a detailed analysis and comparison of the candidate's skills and the skills required by the job description are presented in Fig. 6.

Match percentage: 93%

Matches:

	requirement	explanation
0	Proven software development experience and Android skills development	The candidate has over 6 years of experience in the IT industry, with 4+ years of experience as a Mobile Application Developer. They have experience in developing Android apps using Android Studio and have published multiple apps on the Google Play Store.
1	Proven working experience in Android app development	The candidate has over 4 years of experience as an Android Developer, with experience in developing apps for Android phones, tablets, and wearables.
2	Have published at least one original Android app	The candidate has published multiple Android apps on the Google Play Store.
3	Experience with Android SDK	The candidate has experience in developing Android apps using Android Studio and the Android SDK.
4	Experience working with remote data via REST and	The candidate has experience working with remote data via REST and in their previous roles as an Android Developer.

Missing skills:

	requirement	explanation
0	Experience with third-party libraries and APIs	The candidate's resume does not mention any experience with third-party libraries or APIs.
1	Working knowledge of the general mobile landscape, architectures, trends, and emerging technologies	The candidate's resume does not mention any knowledge of the general mobile landscape, architectures, trends, or emerging technologies.
2	Solid understanding of the full mobile development life cycle	The candidate's resume does not mention any experience with the full mobile development life cycle.

Fig. 5. Resume analysis Case 1.

Match percentage: 30%

Matches:

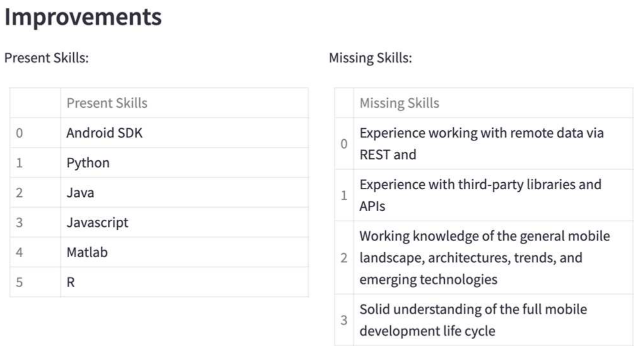
	requirement	explanation
0	Experience with Android SDK	The candidate has experience with Java, which is a programming language used for Android development.
1	Experience working with remote data via REST and	The candidate has experience with Python, which is a programming language that can be used to work with remote data via REST and .
2	Experience with third-party libraries and APIs	The candidate has experience with NumPy, SciPy, and Pandas, which are third-party libraries for Python.
3	Working knowledge of the general mobile landscape, architectures, trends, and emerging technologies	The candidate has experience with mobile app development, which gives them a working knowledge of the general mobile landscape, architectures, trends, and emerging technologies.

Missing skills:

	requirement	explanation
0	Proven software development experience and Android skills development	The candidate has experience with software development, but not specifically with Android development.
1	Proven working experience in Android app development	The candidate has experience with mobile app development, but not specifically with Android app development.
2	Have published at least one original Android app	The candidate has not published any Android apps.
3	Solid understanding of the full mobile development life cycle	The candidate has experience with mobile app development, but not specifically with the full mobile development life cycle.

Fig. 6. Resume analysis Case 2.

Tab 3 Improvements: The Improvements Tab offers valuable insights and recommendations on customizing the resume to align with the job requirements. It highlights relevant skills that the candidate may have underemphasized and identifies areas that may benefit from strengthening. As seen in Fig. 7, this tab presents the relevant skills that a candidate might have inadvertently overlooked in their presentation and also highlights the areas that could benefit from further improvement.



Improvements

Present Skills:

	Present Skills
0	Android SDK
1	Python
2	Java
3	Javascript
4	Matlab
5	R

Missing Skills:

	Missing Skills
0	Experience working with remote data via REST and
1	Experience with third-party libraries and APIs
2	Working knowledge of the general mobile landscape, architectures, trends, and emerging technologies
3	Solid understanding of the full mobile development life cycle

Fig. 7. Improvements.

Tab 4 Resources: The Resources Tab recommends a variety of resources, including books and courses as shown in Fig. 8, which candidates can use to further develop their skills.

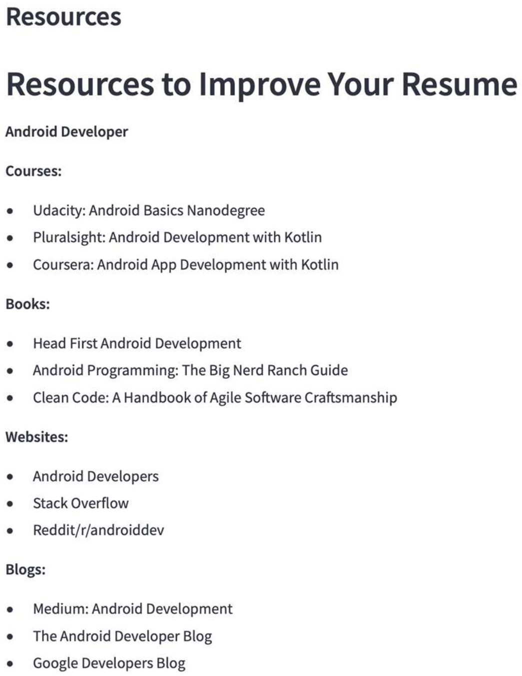


Fig. 8. Resources.

A study was carried out in collaboration with Embassy IT Solutions, a Bangalore-based IT company. The study analyzed 47 resumes using our platform and the industry-standard ATS system used by the HR department. The study revealed that out of 39 technical resumes analyzed, our platform performed better in analyzing 33 resumes while out of 8 non-technical resumes analyzed, our platform did better in analyzing 3 of them. This study indicates that our platform performs better than the industry standard ATS system in analyzing and comparing technical resumes but performs equally well as the industry standard ATS system in analyzing non-technical resumes. This demonstrates the system's ability to scale and adapt to diverse industries and job markets.

VII. Conclusion

In this paper, we delved into the possibilities of utilizing advanced technology such as Large Language Models (LLM) and Natural Language Processing (NLP) to effectively analyze and compare resumes or CVs with job descriptions. Our application uses Gemini Pro LLM, a Google-developed LLM, to process resume and job description data. The primary aim of the application is not only to evaluate and match the user's CV with the job description but also to provide valuable feedback and suggestions on how to improve the CV. With the increasing number of job seekers, companies often receive a flood of resumes, making it difficult to quickly identify the most qualified candidates. Our application makes it possible to extract relevant information from resumes with speed and accuracy, enabling recruiters to streamline the selection process. The future direction of this system involves developing an application that can analyze multiple resumes for a job description simultaneously and generate a comprehensive report with details of all the analyzed resumes. Additionally, there is potential for developing more sophisticated algorithms and models to enhance the accuracy and reliability of resume screeners and ATS systems, particularly with complex and dynamic resume templates. Although some measures have been taken to reduce bias in the prompt analysis module by creating an unbiased System Prompt for the LLM, there is still room for improvement. Future work could involve incorporating additional strategies to mitigate bias within the LLM, thus promoting fairness and inclusivity in the automated screening process. Conducting user studies and evaluations to assess the effectiveness and corporate acceptance of resume screeners in various settings, including corporate workplaces and educational institutions, can provide valuable insights for further development and implementation.

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